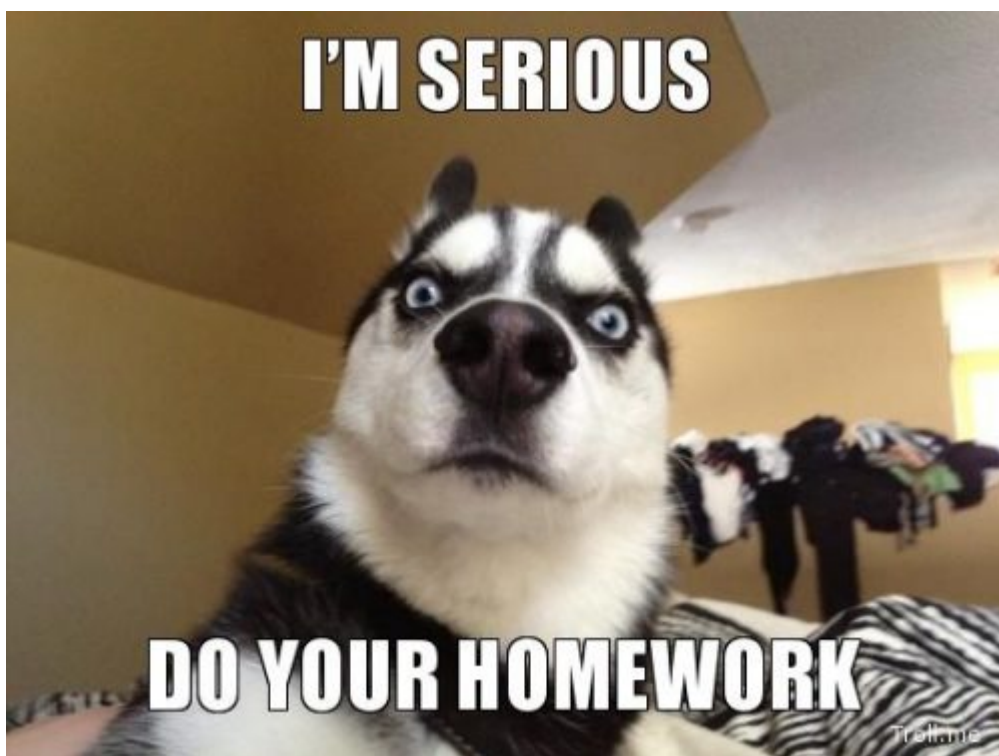


科学课作业（未来7天）

Yupei Duan · December 19, 2016 · SciShine Popular Science Archive



各位同学好，由于雾霾原因，我们要停课3天，另外，周五我要出差（上周五已补课），不能上科学课，再另外，12月26日放圣诞节假一天……所以下次我与大家在科学课上相见的时间是12月27日（周二上周一的课）——博雅班，12月30日（周五）——思贤班。大家至少有7天的时间来完成以下的任务，请合理分配你的时间和精力，期待你们的优秀表现：

1. 教材：阅读教材P27以前所有内容，并完成教材上所有的练习，下次课上检查,记入AMS。

Lesson 2

Lesson Review

Vocabulary

Draw a line to connect the following terms to their definitions.

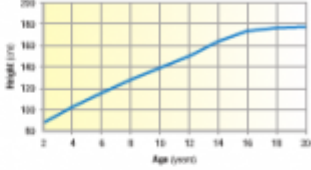
- 1 skeletal system
- 2 ligament
- 3 muscular system
- 4 joint
- 5 tendon

- A groups of muscles that allow you to move and that move materials inside your body
- B a place where two or more bones connect
- C bones, cartilage, and the ligaments that hold bones together
- D tough strands of tissue that connect muscles to bones
- E a type of tough, flexible connective tissue that holds bones together

Critical Thinking

Use this graph to answer the following questions.

Growth Chart of a Boy



- 10 **Analyze** At which points in this graph is bone growing at the fastest rate?

- 11 **Infer** At which times on this graph would you expect that the boy's growth plates have stopped creating new bone?

- 12 **Apply** If aerobic exercise improves heart strength so that it pumps more blood with each beat, what likely happens in the heart rate as the cardiac muscle gets stronger? Explain your answer.

Key Concepts

- 6 **List** What are the functions of the skeletal system?

- 7 **Analyze** What are bones made of?

- 8 **Explain** How do muscles work in pairs to move the body?

- 9 **Identify** What bone disease is caused by a lack of calcium in the diet?

Lesson 2 The Skeletal and Muscular Systems 27

2. 学案：完成周五所发的正反两面的有关肌肉的学案，需要上交纸质版并且评定一次平时成绩（课上交），计入AMS：

5

Name: _____
Date: _____

THE SKELETAL AND MUSCULAR SYSTEMS

The Muscular System

Directions: Use the words from the box at the bottom of the page to fill in the blanks in the following paragraphs. Not all the words in the box will be used in the selection.



The human muscular system is made up of more than _____ muscles. Muscles are attached to bones by strong connective tissue called _____. There are two groups of muscles. Muscles that we can control with conscious thought are called _____ muscles. The other group of muscles are automatic. We don't have to think about their action. Muscles that move food through our digestive system or keep our hearts beating are _____ muscles.

There are three kinds of muscles found in the human body. Voluntary muscles attached to bones and capable of permitting body movement are called _____ muscles. Involuntary muscles found inside organs such as the stomach, intestines, and blood vessels are called _____ muscles. The third type of muscle is the hardest working muscle and is an involuntary muscle called _____ muscle.

Voluntary muscles receive the signal to contract or relax from the brain. People make the decision to make a movement and the signal is sent from the brain down through the _____ and to the appropriate muscles. When the muscle receives the message to contract or relax, it does so completely. This means that there is no such thing as a partial contraction. The strength or weakness of muscle contractions is determined by the number of muscle _____ involved.

tendon ligaments voluntary involuntary 200 600 tissue skeletal smooth cardiac triceps biceps spinal column budge contract fibers fracture cramp

7

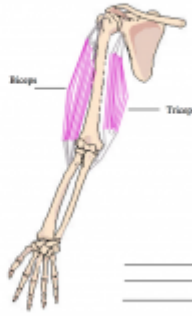
Name: _____
Date: _____

THE SKELETAL AND MUSCULAR SYSTEMS

The Way Muscles Work

Muscles work in pairs to move bones and provide the skeletal system with motion. A good example of the way this works is found in the arm. When we bend our arms up and down, the triceps and biceps muscles are working together to perform the movement.

Directions: Describe how the biceps and triceps work together to straighten and bend the arm.

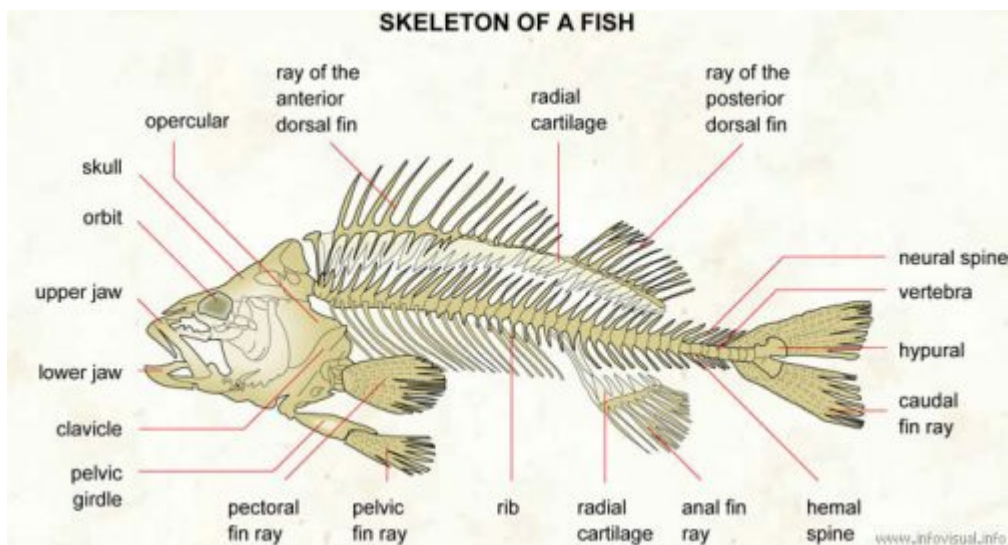


Bending Arm

Straightening Arm

3. 实验：由表及里地吃一条鱼（或一个鸡腿），列出你发现的鱼的骨骼系统结构（或者鸡腿的肌肉、骨骼系统结构），并且拍照、记录，做成示意图（电子版，或摆在纸上用笔标出相应结构再拍照），发送到科学课微信群中（Deadline:12月26日20:00）。

例1:



(参考：鱼的骨骼系统结构示意图)

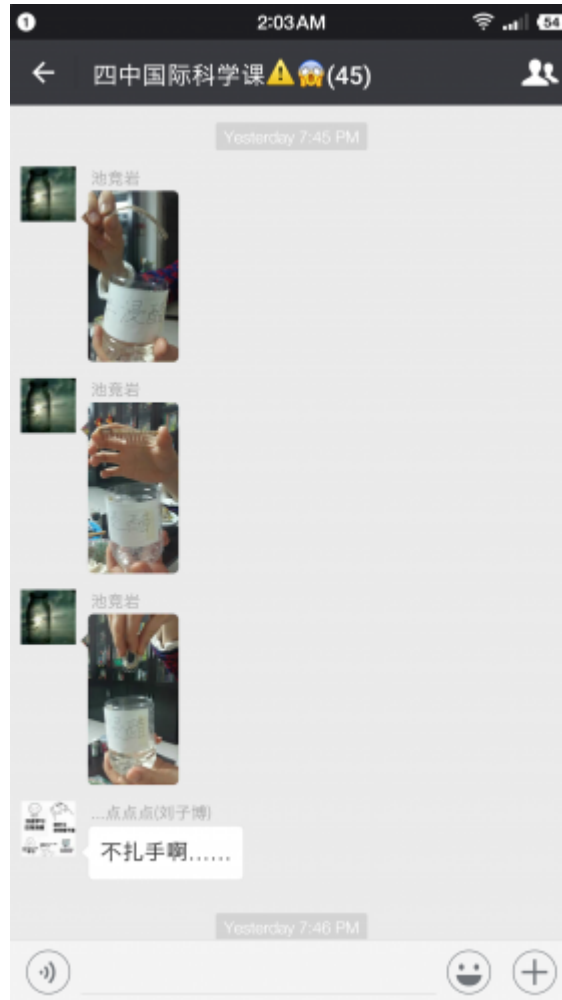
例2:



(参考：鸡翅结构示意图)

4. 实验：将一根鱼刺（或一根鸡腿骨）打一个结，将照片发送到科学课微信群中（Deadline:12月26日 20:00）。

以池竞岩同学作业为例：



实验参考源文件下载: [Bend A Bone With Vinegar1](#) (PDF)

5. 选做加分: 观看影片 *Fantastic Voyage* (90mins)。从潜艇进入人体的结构开始, 到离开人体的结构结束, 整理潜艇在人体中的探险路线, 将影片中所展示出的潜艇所经过的人体内部各个主要结构进行截图, 并用文字进行简要说明, 并进行简单的自我评述。将电子版文件发送到

Danny.Duan@bhsfc.com(Deadline:2016年12月26日13:00),文件名: Fantastic+学号+姓名



影片介绍: https://en.wikipedia.org/wiki/Fantastic_Voyage

影片欣赏: <https://www.youtube.com/watch?v=eltKPv-n9sk>

(观看视频需要VPN)

有任何问题请先用搜索引擎解决。

6. 选做加分: 观看影片 *Osmosis Jones* (90mins) , 简要评述人体的免疫系统是如何抵抗有害病毒和细菌的。将电子版文件发送到 Danny.Duan@bhsfic.com (Deadline:2016年12月26日13:00) ,文件名:
Osmosis+学号+姓名



影片介绍: https://en.wikipedia.org/wiki/Osmosis_Jones

影片欣赏: <https://www.youtube.com/watch?v=tbKlcoss6LA>

(观看视频需要VPN)